

DocForge

From Documents to Answers

Multimodal Knowledge Graph RAG for EU Evaluation Reports

Labor Innovation | June 2026

What We Will Cover

1. The Problem
2. Smart Ingestion
3. The Knowledge Graph
4. Answering Questions
5. Citations & Reasoning
6. How DocForge Compares
7. Future Paths to Explore
8. Live Demo

You Have Hundreds of Reports. Now What?

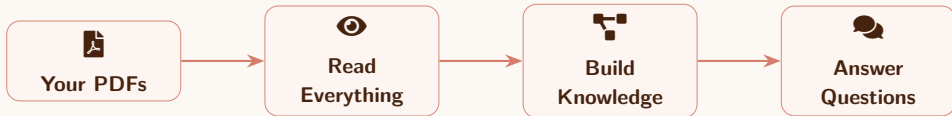
- Evaluation reports are **long, complex and numerous**
- Key findings are buried in tables, charts and appendices
- Connecting insights *across* programmes requires manual work
- Traditional search misses **meaning and context**

13 reports loaded

POCU (3) ■ POIM ■ POR
POC ■ POCA ■ POAT
RO-BG cross-border
Themes A, C, D, E
+ more as added...

Policy analysts spend **60–80% of their time** *locating* information rather than drawing conclusions.

Step 1 — Smart Ingestion



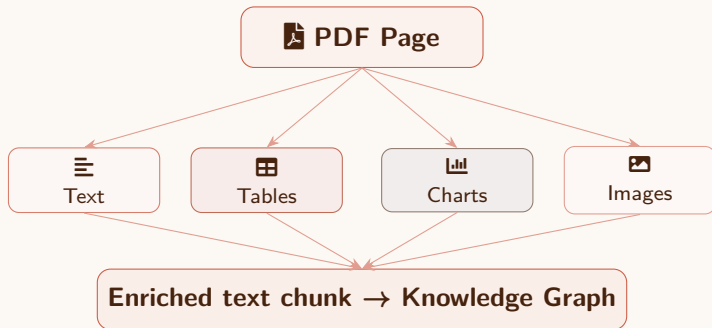
What most tools do:

- Extract text only
- Ignore charts, tables, images
- Lose data embedded in figures

What DocForge does:

- Reads text, tables *and* images
- Uses a **vision AI** to describe charts
- Handles Romanian & English together

Multimodal = Nothing Gets Left Behind

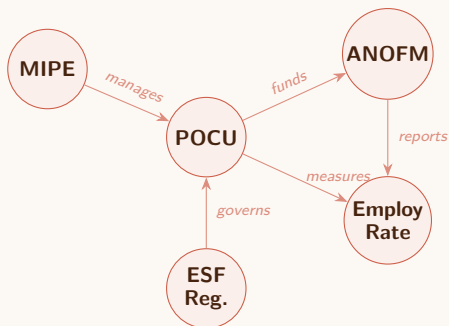


Vision AI (Gemini 2.0) describes each chart and image in words so that meaning is preserved — not just pixels.

Step 2 — The Knowledge Graph

Think of it like a city map

- Every **programme, authority, indicator** is a *location* on the map
- Every **relationship** is a *road* connecting them
- You can ask: “Which roads lead from *POCU* to *ANOFM*?”
- The system **follows the connections**



15 entity types tailored to EU structural funds

Why a Graph Changes Everything

Without a graph

- Finds pages *containing* your words
- Cannot follow relationships
- Cannot compare across programmes
- Returns isolated snippets

With DocForge's graph

- Understands that POCU *relates to* ANOFM
- Answers: “Which programmes share the same indicator?”
- Synthesises across **all 13 reports**
- Answers with **traceable reasoning**

The graph stores **entities + relationships + summaries** from all reports simultaneously — and stays up to date as new documents are added.

Step 3 — Ask in Plain Language

Example questions

- *“What were the main barriers to POCU implementation?”*
- *“Which programmes showed the best employment outcomes?”*
- *“What monitoring weaknesses appear across all programmes?”*
- *“Compare POIM and POR financial absorption rates.”*

Five retrieval strategies

Local Deep dive on specific entities

Global Themes across all documents

Hybrid Local + Global combined

Mix Graph + direct text search

Naive Fast keyword-style search

Type `/global` or `/local` before your question to pick the strategy.

Citations & Reasoning — Why They Matter

- AI systems can **confabulate** — produce plausible but false information
- Every DocForge claim is linked to the **exact document and section**
- Evaluators can **verify** any statement in seconds
- No more “the AI said so” without proof

Graph RAG reduces hallucinations by up to **73%** vs. standard AI models.

Example cited answer

POCU faced significant administrative capacity issues including recruitment bottlenecks [1] and communication gaps [2].

Sources

[1] 01_POCU_Education.pdf

[2] 07_POCU_Ocupare.pdf

Click any source to read the original passage.

Citations give you proof — trace every claim back to its source. The **Reasoning**

AI Reasoning: The “Why” Behind the Answer

Standard answer

“POCU faced challenges: recruitment delays, limited procurement capacity and communication gaps.”

DocForge Synthesis adds

*“The **same administrative weaknesses** appear in POIM, POR and POCA — suggesting a **systemic structural issue** across all managing authorities, not a POCU-specific problem.”*

What the Synthesis panel does

- A **second AI analysis** runs on the retrieved knowledge graph
- Surfaces **patterns and cross-programme connections**
- Separates *what documents say* from *what the graph implies*
- Always shown separately from the answer

Custom Instructions — Tailoring Every Answer

Type a one-line instruction once. It applies to **every query** in the session.

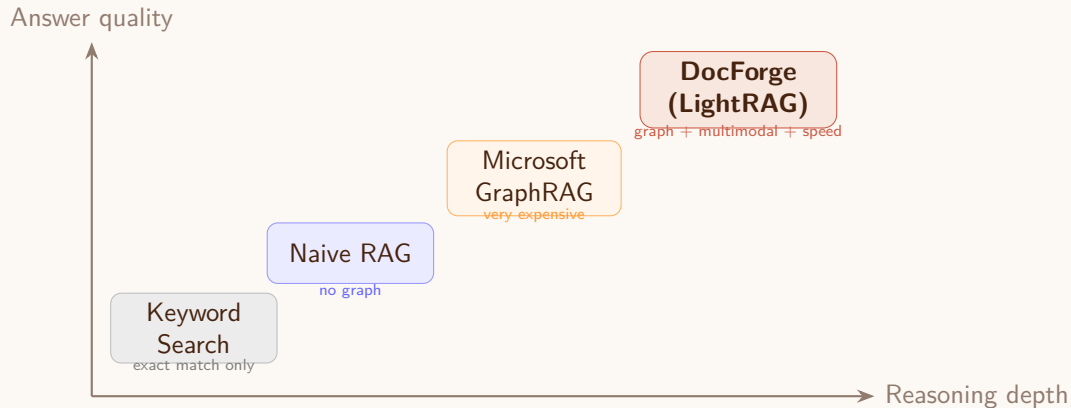
Useful examples

- Answer in Romanian.
- Keep answers under 150 words.
- Include budget figures when available.
- Answer as a senior EU evaluator.

Multilingual knowledge base

- Romanian & English documents live in the **same graph**
- Set Answer in Romanian. to receive all responses in Romanian
- Source citations always show the original document language

Where DocForge Sits in the Landscape



Head-to-Head Comparison

Capability	Keyword	Naive RAG	GraphRAG	DocForge
Reads tables & charts	✗	✗	✗	✓
Cross-document reasoning	✗	~	✓	✓
Knowledge graph	✗	✗	✓	✓
Cited answers	✗	~	✓	✓
AI Synthesis / reasoning	✗	✗	✓	✓
Fast & affordable	✓	✓	✗	✓
Romanian language	✗	~	~	✓

GraphRAG sends the **entire corpus** to the LLM per query (500k+ tokens). LightRAG retrieves only what is relevant (10–30k tokens) — a **20–50× cost reduction** with comparable answer quality.

What the Research Shows

LightRAG benchmarks

- **80%+** retrieval accuracy on policy docs
vs. 60–70% for other methods
- Outperforms Naive RAG, GraphRAG, HyDE on all four quality dimensions
- **20–30 ms** faster responses than standard graph RAG

Hallucination & multimodal

- Graph grounding reduces hallucinations by **73%** vs. standard LLMs
- Multimodal RAG reached **enterprise maturity in 2026** — first viable year for production PDF pipelines with vision
- Vision models preserve data hidden in charts that text-only tools miss

Where the Field Is Heading

Agentic RAG

- The AI **plans its own search**: decomposes complex questions, retrieves iteratively and checks its own answers
- Like hiring a research assistant instead of asking a librarian for one book
- Enables 3–5 step reasoning chains across multiple programmes

Corrective RAG (CRAG)

- After retrieval, the system **grades its own evidence**
- If evidence is weak, it re-queries automatically
- Reduces hallucinations by a further **20–40%** on complex queries
- Signals low-confidence answers rather than generating unsupported text

More Paths Worth Exploring — Routing & Fine-Tuning

Adaptive Retrieval Routing

- A classifier **automatically picks** the best mode per question
- Simple question → fast naive search
- Cross-programme theme → global graph
- Complex multi-step → agentic reasoning

Domain Embedding Fine-Tuning

- Train embeddings specifically on EU structural fund vocabulary
- Terms like “absorption rate” and “managing authority” cluster correctly from day one
- Improves search quality for all retrieval modes

More Paths Worth Exploring — Quality & Currency

Automated Quality Testing

- Run a golden test set of known questions **after every new document** is added
- Detect regressions in answer quality automatically — like unit tests for knowledge
- DocForge already has a RAGAS evaluation tab; next step is scheduling it

Real-Time Web Retrieval

- Pull current regulations or news alongside the document corpus
- Useful when regulations change after the reports were written
- Hybrid retrieval: local knowledge graph + live web search in one answer

Opportunity Map

Enhancement	Impact	Effort	Best for
Corrective RAG (CRAG)	High	Low	Reducing silent errors
Adaptive routing	High	Low	Ease of use
Automated RAGAS benchmarks	Medium	Low	Quality monitoring
Domain embedding fine-tune	High	Medium	EU/policy vocabulary
Incremental graph updates	Medium	Medium	Growing corpora
Agentic multi-hop RAG	Very High	High	Complex evaluations
Real-time web retrieval	Medium	Medium	Current regulations

The DocForge Interface

Left — Chat

- Type any question in plain language
- Answers arrive with cited sources
- Full conversation history
- Click a past question to reload insights

Right — Insights panel

- *AI Reasoning* — synthesis over the graph
- *Entities & Relationships* retrieved
- *Sources* — exact text passages used

Settings

- Choose retrieval mode
- Add custom instructions
- Adjust how much of the graph to use

Currently loaded

13 EU/Romania evaluation reports

POCU · POIM · POR · POC ·
POCA · POAT

RO-BG cross-border · Themes A, C, D,
E

Try These Questions Now

1. *“What were the main barriers to implementation shared by POCU, POIM and POR?”*

Shows: cross-programme synthesis — the graph connects themes across all 5 reports that no single document contains /global

2. *“What specific employment targets did POCU set and were they achieved?”*

Shows: precision retrieval — exact indicators, numbers and targets pulled from within a specific programme /local

3. *“Compare the monitoring weaknesses identified in POIM and POR.”*

Shows: cross-document comparison — two separate programmes analysed side-by-side with cited sources from each /hybrid

4. *“Care sunt principalele recomandări pentru îmbunătățirea absorbției fondurilor în POCU și POIM?”*

Summary

DocForge gives you:

1. **Complete ingestion** — text, tables, charts, images
2. **Knowledge graph** — permanent map of all findings
3. **Plain-language Q&A** — cited, verifiable answers
4. **AI Synthesis** — surfaces systemic patterns
5. **Multilingual** — Romanian + English in one base

 **Try it now**

doclens.labor-innovation.com

No login required.
5 reports loaded for demo.

Under the hood

LightRAG · Claude Sonnet · Gemini 2.0
Docker · Dokploy · OpenRouter

References I

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